



Lessons Learned Report

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Contents

1	General summary of the project	2
2	Fulfillment of our goals	2
2.1	Follow-up of objectives	2
2.1.1	List of obtained goals	2
3	What we learned (main part)	3
3.1	Good Experiences	3
3.1.1	Successful Interface Integration	3
3.2	Bad Experiences	3
3.2.1	Difficulty in Project Initiation and Problem Understanding	3
4	Conclusion	4

1. General summary of the project

This project focused on the implementation and verification of a parallel and a bit-serial version of a RISC-V processor. Both processor models needed to be implemented on one FPGA and tested by an advanced testbench implemented on another FPGA, with the CPU and testbench communicating through physical wires. This setup is intended to operate in the extreme environments of Venus, specifically under high-temperature conditions. The work uses KTH's experimental Silicon-on-Insulator technology, which can operate at temperatures approaching Venus's 480°C requirement. However, the test station has a constraint of only 12 pads available, requiring the development of a serialization system and communication protocol.

The project was carried out by dividing the work into two main sub-projects: the testbench development with test set and implementation, and the communication system with serialization and protocol for data transfer. This separation enabled parallel development and allowed team members to focus on different topics and architectures, ultimately achieving better efficiency. The team was organized into two subgroups: three members worked on the testbench (functional testing, CoreMark generation, and integration with RAM and UART), while three members focused on the bit-serial interface (frequency and pin properties, communication protocol specification, and RTL implementation). A significant portion of the work involved learning and adapting to unfamiliar tools, technologies, and design constraints, which strongly influenced both the schedule and working methods.

Regarding planned versus actual outcomes, the project timeline remained largely on track despite several technical challenges. The buffer time allocated during initial planning proved valuable in absorbing unforeseen delays. From a budget perspective, as an academic project no formal budget was involved, though minor unplanned expenses occurred for purchasing power adapters and communication cables between FPGA boards, which team members handled individually. In terms of results, the project successfully delivered a functional dual-FPGA system with a CPU and testbench connected by bit-serial communication. The test for CISC-V and testbench across 2 FPGAs runs successfully with measured performance at 100,198 IPS approaching the 1MHz clock of the CISC-V. The RISC-V has also been tested with the testbench, but only in simulation without the bit-serial communication.

Several significant problems impacted the project timeline and workflow. The initial codebase delivery was delayed, and upon receipt contained various compilation errors and bugs that were assumed to be outside the project scope and already corrected. This required the team to spend considerable time debugging and refactoring code that should have been functional. Additionally, some FPGA boards malfunctioned and needed replacement, and students had to purchase their own adapters and cables for inter-board communication. While these hardware issues were relatively minor and easily resolved, they introduced delays in the integration phase. The Christmas holiday period presented organizational challenges, as not all team members could work together in person. This necessitated a division of tasks between those working remotely and those with physical access to the two FPGA boards required for final integration. Since each member initially had one board, those who traveled left their boards with team members remaining on campus to enable dual-board testing to continue. The holiday period also reduced overall team availability, requiring careful task prioritization and remote workflow coordination. Despite these obstacles, the project maintained progress through adaptive planning, clear task division, and effective cross-team coordination.

2. Fulfillment of our goals

2.1. Follow-up of objectives

The project successfully achieved the majority of its stated technical and business objectives, demonstrating a functional dual-FPGA validation platform despite encountering significant implementation challenges.

2.1.1 List of obtained goals

The following table summarizes the achievement status of all project objectives defined in the initial planning phase:

ID	Objective	Status	Notes
T1	Validation of both processor architectures using a suite of at least 10 assembly-level test programs to verify core ISA functionality	Partially achieved	Both processors validated in simulation; only CISC-V validated on hardware
T2	Execution and verification of a complex benchmark algorithm (CoreMark), with results transmitted to the host PC	Achieved	Successfully executed in simulation for both processors
T3	Demonstration of stable data transfer exceeding 1,000 instructions without corruption between the two FPGAs	Achieved	Verified during CISC-V hardware testing with measured performance of 100,198 IPS
T4	(Optional) Collection of quantitative performance metrics such as maximum clock frequency, benchmark cycle counts, and FPGA resource utilization	Partially achieved	Maximum clock frequency measured; comprehensive resource analysis incomplete
B1	Complete all design, implementation, and validation phases within the planned 16-week schedule	Achieved	All major deadlines met despite technical challenges
B2	Strengthen teamwork, communication, and project management skills	Achieved	Effective collaboration and adaptive planning throughout the project
B3	Deliver a high-quality, fully functional system and comprehensive documentation	Achieved	Functional dual-FPGA system delivered with documentation

Table 1: Summary of project objectives and achievement status

The primary unfulfilled objective was the hardware validation of the RISC-V processor on the dual-FPGA platform. While the RISC-V design passed all functional tests in simulation and was verified to implement the instruction set correctly, integration with the bit-serial communication interface was not completed. This limitation stemmed directly from discrepancies in the initial codebase delivery. The project planning assumed both processor implementations would be provided as consistent black-box modules with identical external interfaces, enabling straightforward integration with the testbench through a common protocol. However, the delivered implementations featured different interface specifications, requiring significant additional integration effort that could not be accommodated within the project timeline after addressing codebase debugging needs.

The successful completion of the CISC-V validation demonstrates that the project methodology is sound and that the remaining objectives are achievable with focused effort on interface standardization and systematic performance characterization.

3. What we learned (main part)

All along the project, the team encountered both positive and less positive experiences. It significantly contributed to our learning, from a technical point of view but also organizational one. This section highlights five aspects that worked well and five that were more difficult, along with an explanation of why these outcomes occurred.

3.1. Good Experiences

3.1.1 Successful Interface Integration

One of the most rewarding experiences of the project was the moment when individual components were finally integrated together. The final integration was more complex and time-consuming than initially expected. So it is even more satisfying to see 2 interfaces of different people in the group join together. Seeing the result of our work is nice and it confirmed that the architectural decisions, communication protocols, and verification strategies were good choices.

3.2. Bad Experiences

3.2.1 Difficulty in Project Initiation and Problem Understanding

A major challenge at the beginning of the project was the difficulty in clearly understanding the scope of the work and how to effectively start the implementation. Indeed, the project continued previous work and we had to understand what was already done and what needed to be done. Moreover, the project had existing designs and architectures that were not documented. This led to inefficiencies at the beginning. As a result, a lot of time was spent clarifying objectives, understanding

code, and identifying components. This experience highlighted the importance of having a good general overview of the project before starting implementation. It can be really challenging to transition from theoretical knowledge to practice on a real project.

4. Conclusion